What is Data Mining?

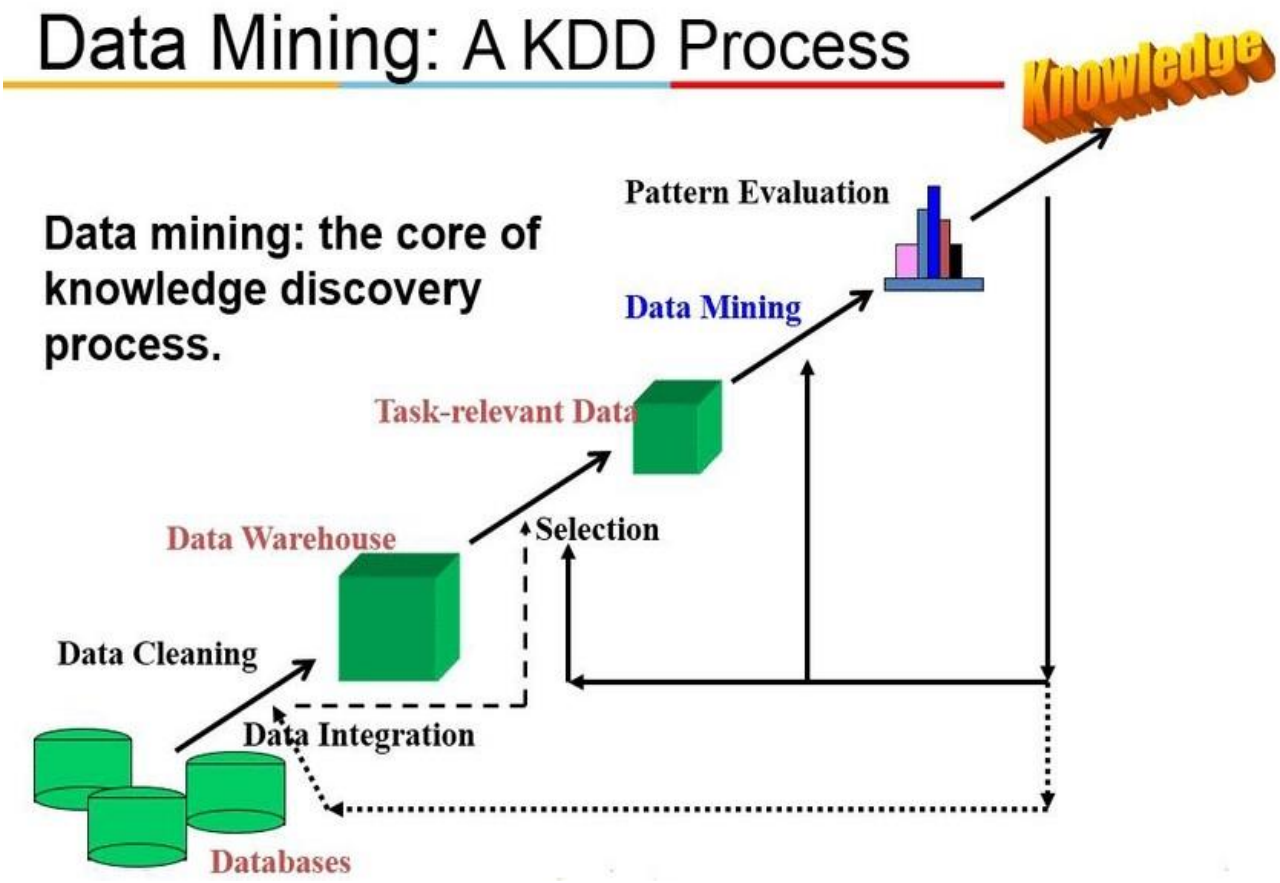
Data Mining is defined as extracting information from huge sets of data. In other words, we can say that data mining is the procedure of mining knowledge from data. The information or knowledge extracted so can be used for any of the following applications

- Market Analysis
- Fraud Detection
- Customer Retention
- Production Control
- Science Exploration

What is Knowledge Discovery?

Some people don't differentiate data mining from knowledge discovery while others view data mining as an essential step in the process of knowledge discovery. Here is the list of steps involved in the knowledge discovery process –

- **Data Cleaning** − In this step, the noise and inconsistent data is removed.
- **Data Integration** − In this step, multiple data sources are combined.
- **Data Selection** − In this step, data relevant to the analysis task are retrieved from the database.
- **Data Transformation** − In this step, data is transformed or consolidated into forms appropriate for mining by performing summary or aggregation operations.
- **Data Mining** − In this step, intelligent methods are applied in order to extract data patterns.
- **Pattern Evaluation** − In this step, data patterns are evaluated.
- **Knowledge Presentation** − In this step, knowledge is represented.



What Is Data Mining?

Data Mining is in big demand today as it helps the businesses study how the sales of their products can increase. We can understand this by an example of a fashion store, which will register each of its customer who purchases an item from their store.

Based on the data given by the customer such as age, gender, income group, profession, etc., the store will be able to find out which type of customers buy different products. Here, we can see that the name of the customer is of no use as we cannot predict the trend of purchase by name as to whether that person will buy a certain product or not.

Thus the useful information can be found out using the age group, gender, income group, profession, etc. Searching for knowledge or interesting pattern in data is “Data Mining”. Other terms that can be used in place are Knowledge Mining from data, Knowledge Extraction, Data Analysis, Pattern Analysis, etc.

Another term that is popularly used in data mining is Knowledge Discovery from Data or

KDD.

Data Analysis Process

The knowledge discovery process is a sequence of the following steps:

1. **Data Cleaning:** This step removes noise and inconsistent data from the input data.

2. **Data Integration:** This step combines multiple sources of data. The data cleaning and data integration step together to form the preprocessing of data. The preprocessed data is then stored in the data warehouse.

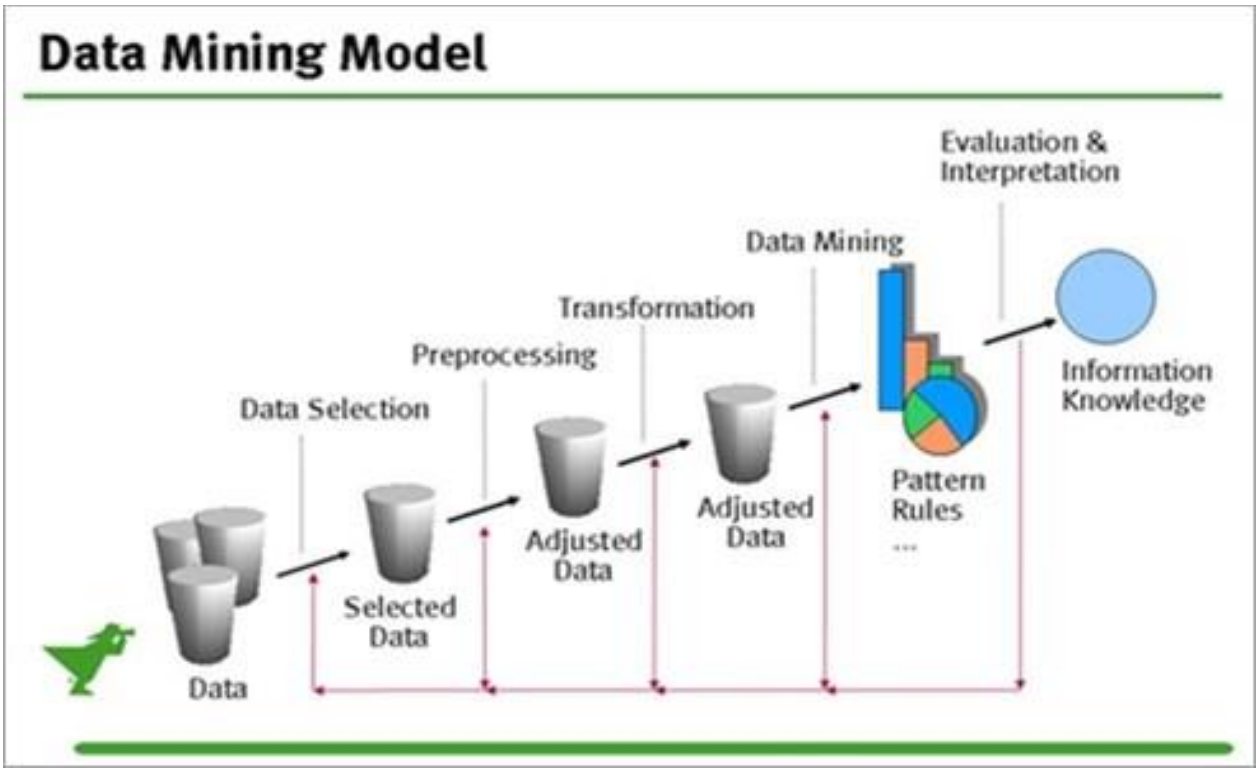
3. **Data Selection:** These steps select the data to the analysis task from the database.

4. **Data Transformation:** In this step, various data aggregation and data summary techniques are applied to transform the data into a useful form for mining.

5. **Data Mining:** In this step, data patterns are extracted by applying intelligent methods.

6. **Pattern Evaluation:** The extracted data patterns are evaluated and recognized according to the interestingness measures.

7. **Knowledge Representation:** Visualization and knowledge representation techniques are used to present the mined knowledge to the users.



The steps 1 to 4 come under the data preprocessing stage. Here, data mining is represented as a single step but it refers to the entire knowledge discovery process.

Thus, we can say, that data analysis is the process of discovering interesting patterns and knowledge from a large amount of data. The data sources can include databases, data warehouses, World Wide Web, flat files and other informative files.

What Kinds Of Data Can Be Mined?

The most basic forms of data for mining are database data, data warehouse data, and transactional data. The data mining techniques can also be applied to other forms like data streams, sequenced data, text data, and spatial data.

#1) Database Data: The database management system is a set of interrelated data and a set of software programs to manage and access the data. The relational database system is a collection of tables and each table consists of a set of attributes and tuples.
Mining of relational databases search the trends and data patterns **E.g**. credit risk of customers based on age, income, and previous credit risk. Also, mining can find out deviations from the expected **E.g.** a significant increase in the price of an item.

#2) Data Warehouse Data: A data warehouse is a collection of information collected from multiple data sources, stored under a unified schema at a single sit. A DW is modeled as a multidimensional data structure called data cube having cells and dimensions providing precomputation and faster access to data.

Data mining is performed in an OLAP style by combining the dimensions at varying levels of granularity.

#3) Transactional Data: Transactional Data captures a transaction. It has a transaction id and a list of items used in transaction.

#4) Other kinds of Data: Other data can include: time-related data, spatial data, hypertext data, and multimedia data.

What Techniques Are Used In Data Mining?

Data Mining is a highly application-driven domain. Many techniques such as statistics, machine learning, pattern recognition, information retrieval, visualization, etc., influence the development of data analysis methods.

Let's discuss some of them here!!

Statistics

The study of collection, analysis, interpretation, and presentation of data can be done using Statistical Models. **For example**, statistics can be used to model noise and missing data, and then this model can be used in large data set to identify the noise and missing values in data.

Machine Learning

ML is used to improve performance based on data. The main research area is for computer programs to automatically learn to recognize complex patterns and make intelligent decisions based on the data.

Machine Learning focuses on accuracy and data mining focuses on the efficiency and scalability of mining methods on the large data set, complex data, etc.

Machine learning is of three types:

1. **Supervised Learning:** The target data set is known and the machine is trained according to the target values.

2. **Unsupervised Learning:** The target values are not known and the machines learn by themselves.

3. **Semi-Supervised Learning:** It uses both the techniques of supervised and unsupervised learning.

Information Retrieval (IR)

It is the science of searching for documents or information in documents.

It uses two principles:

- Data that is to be searched is unstructured.
- The queries are formed mainly by keywords.

By using data analysis and IR, we can find major topics in the collection of documents and also the major topics involved in each document.

Major Issues In Data Analysis

Data Mining has a number of issues related to it as mentioned below:

Mining Methodology

- As there are diverse applications, new mining tasks continue to emerge. These tasks can use the same database in different ways and require the development of new data mining techniques.
- While searching for knowledge in large datasets, we need to explore multidimensional space. To find interesting patterns, various combinations of dimensions need to be applied.
- Uncertain, noisy and incomplete data can sometimes lead to erroneous derivation.

User Interaction

- The data analyzing process should be highly interactive. It is important for facilitating the mining process to be user interactive.
- The domain knowledge, background knowledge, constraints, etc., should all be incorporated in the data mining process.
- The knowledge discovered by mining the data should be usable for humans. The system should adopt an expressive representation of knowledge, user-friendly visualization techniques, etc.

Efficiency And Scalability

- Data mining algorithms should be efficient and scalable to effectively extract interesting data from a huge amount of data in the data repositories.
- Wide distribution of data, complexity in computation motivates the development of

parallel and distributed data-intensive algorithms.

Diversity of Database Types

- The construction of effective and efficient data analysis tools for diverse applications, wide spectrum of data types from unstructured data, temporal data, hypertext, multimedia data, and software program code remains a challenging and active area of research.

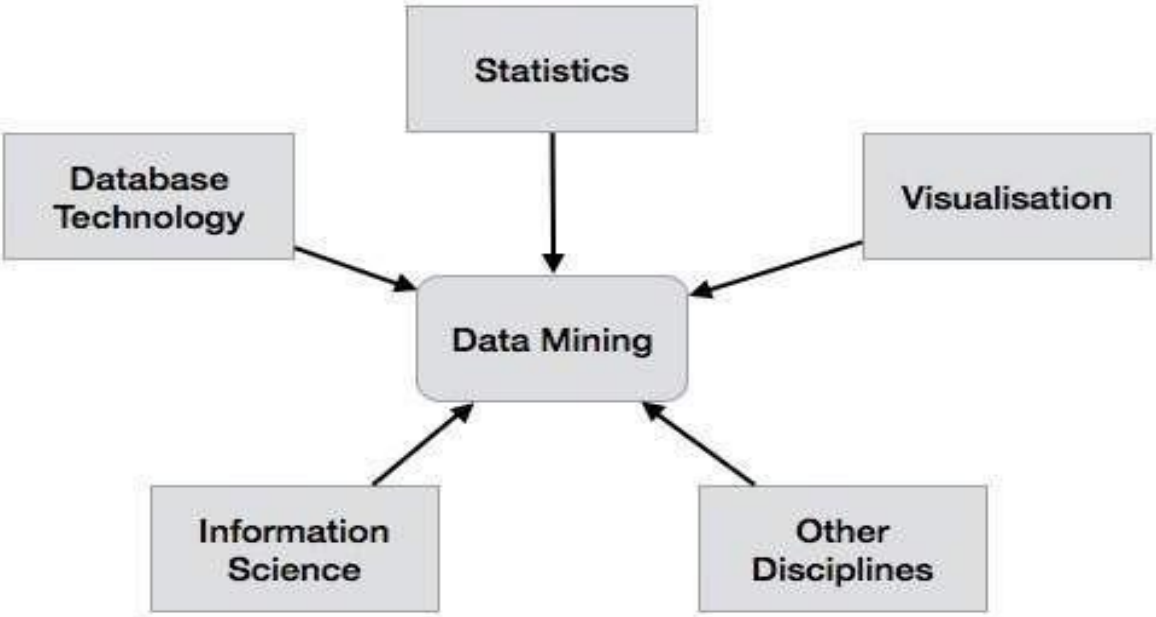
Social Impact

- The disclosure to use the data and the potential violation of individual privacy and protection of rights are the areas of concern that need to be addressed.

Data Mining System Classification

A data mining system can be classified according to the following criteria −

- Database Technology
- Statistics
- Machine Learning
- Information Science
- Visualization
- Other Disciplines



Apart from these, a data mining system can also be classified based on the kind of (a)

databases mined, (b) knowledge mined, (c) techniques utilized, and (d) applications adapted.

Classification Based on the Databases Mined

We can classify a data mining system according to the kind of databases mined. Database system can be classified according to different criteria such as data models, types of data, etc. And the data mining system can be classified accordingly.

For example, if we classify a database according to the data model, then we may have a relational, transactional, object-relational, or data warehouse mining system.

Classification Based on the kind of Knowledge Mined

We can classify a data mining system according to the kind of knowledge mined. It means the data mining system is classified on the basis of functionalities such as −

- Characterization
- Discrimination
- Association and Correlation Analysis
- Classification
- Prediction
- Outlier Analysis
- Evolution Analysis

Classification Based on the Techniques Utilized

We can classify a data mining system according to the kind of techniques used. We can describe these techniques according to the degree of user interaction involved or the methods of analysis employed.

Classification Based on the Applications Adapted

We can classify a data mining system according to the applications adapted. These applications are as follows −

- Finance
- Telecommunications
- DNA
- Stock Markets
- E-mail

Integrating a Data Mining System with a DB/DW System

If a data mining system is not integrated with a database or a data warehouse system, then there will be no system to communicate with. This scheme is known as the non-coupling scheme. In this scheme, the main focus is on data mining design and on developing efficient and effective algorithms for mining the available data sets.

The list of Integration Schemes is as follows −

- **No Coupling** − In this scheme, the data mining system does not utilize any of the database or data warehouse functions. It fetches the data from a particular source and processes that data using some data mining algorithms. The data mining result is stored in another file.
- **Loose Coupling** − In this scheme, the data mining system may use some of the functions of database and data warehouse system. It fetches the data from the data respiratory managed by these systems and performs data mining on that data. It then stores the mining result either in a file or in a designated place in a database or in a data warehouse.
- **Semi−tight Coupling** − In this scheme, the data mining system is linked with a database or a data warehouse system and in addition to that, efficient implementations of a few data mining primitives can be provided in the database.
- **Tight coupling** − In this coupling scheme, the data mining system is smoothly integrated into the database or data warehouse system. The data mining subsystem is treated as one functional component of an information system.

Data Mining Tools

For quicker analysis of data, it is important to use the proper tool for the requirement. Following are the few essential tools used for Data Mining

Rapid Miner – It a very popular open source software that requires no programming to operate. It provides multifaceted data functions such as data pre-processing, predictive analysis and visualization.

Weka – It is a set of machine learning algorithms for data mining. They are either applied directly or called through a Java code. This tool is used to perform data pr-processing, clustering, regression, visualization and classification in data mining.

Orange – It is a Python library that powers Python scripts with Machine Learning and Data Mining algorithms. It is used for classifying, pre-processing, modelling, clustering and other miscellaneous functions.

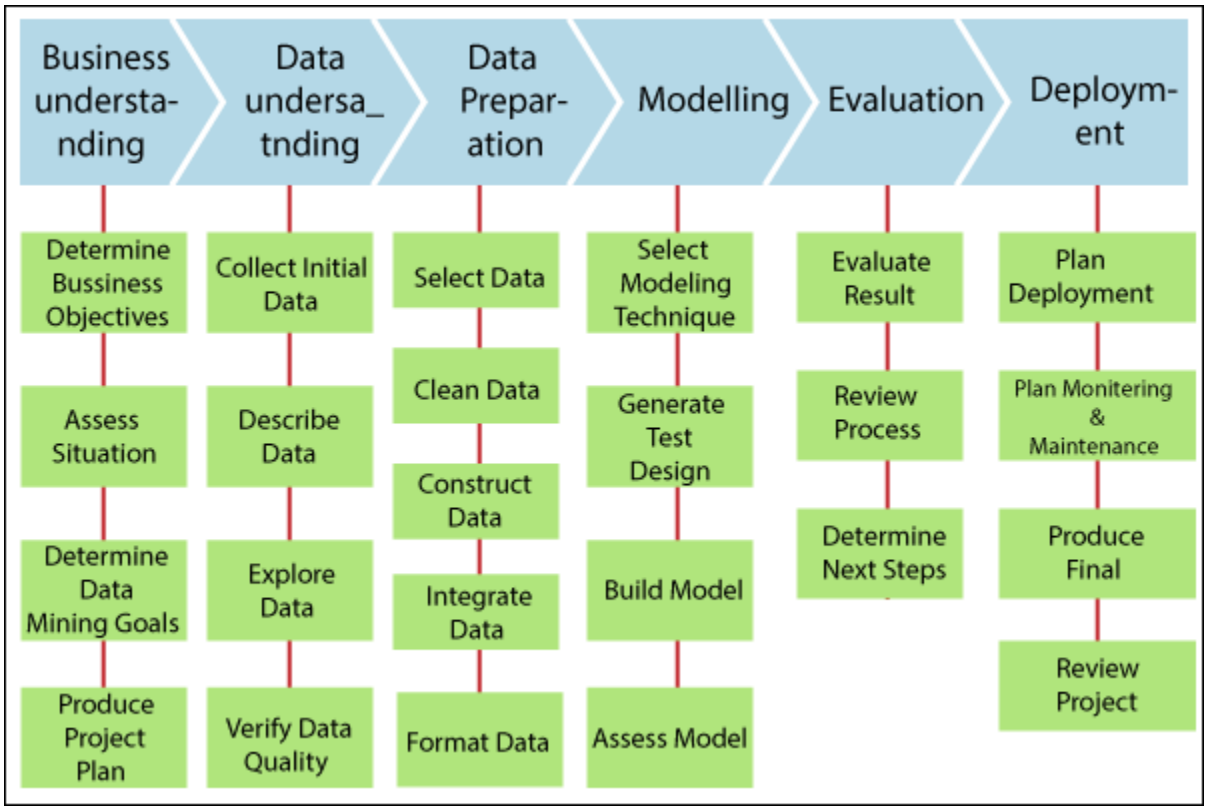
R – It is an opensource software environment widely used for data mining tasks. It comes with huge community support as well as hundreds of libraries specifically built for data mining.

Knime – This tool is primarily used for data pre-processing in data mining. Data pre-processing means extraction, transformation and loading of data. Knime is very popular among the financial data analysts.

Data Mining Implementation Process

Many different sectors are taking advantage of data mining to boost their business efficiency, including manufacturing, chemical, marketing, aerospace, etc. Therefore, the need for a conventional data mining process improved effectively. Data mining techniques must be reliable, repeatable by company individuals with little or no knowledge of the data mining context. As a result, a cross-industry standard process for data mining (CRISP-DM) was first introduced in 1990, after going through many workshops, and contribution for more than 300 organizations.

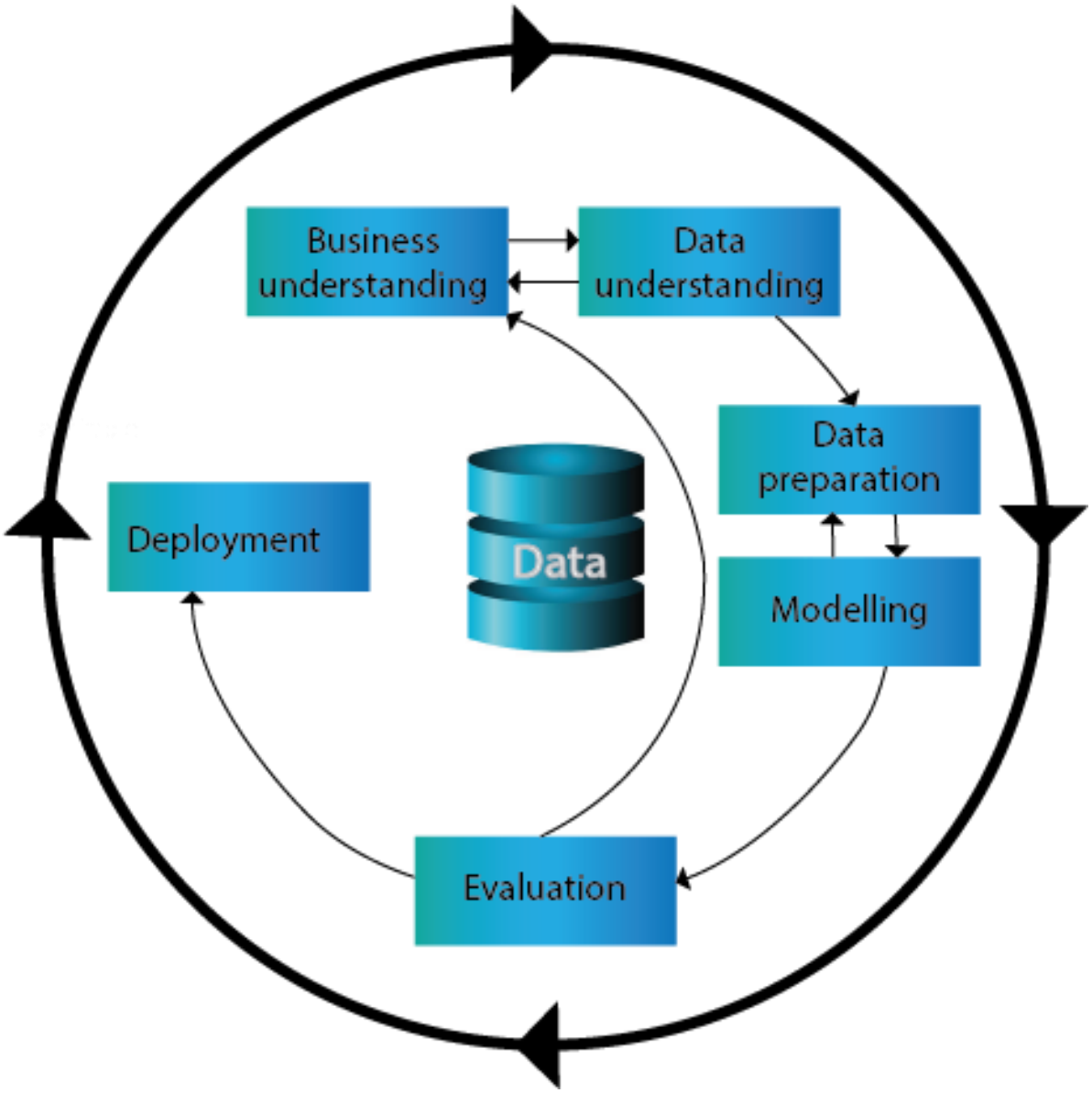
Data mining is described as a process of finding hidden precious data by evaluating the huge quantity of information stored in data warehouses, using multiple data mining techniques such as Artificial Intelligence (AI), Machine learning and statistics.



Let's examine the implementation process for data mining in details:

The Cross-Industry Standard Process for Data Mining (CRISP-DM)

Cross-industry Standard Process of Data Mining (CRISP-DM) comprises of six phases designed as a cyclical method as the given figure:



1. Business understanding:

It focuses on understanding the project goals and requirements form a business point of view, then converting this information into a data mining problem afterward a preliminary plan designed to accomplish the target.

Tasks:

- Determine business objectives
- Access situation
- Determine data mining goals

- Produce a project plan

Determine business objectives:

- It Understands the project targets and prerequisites from a business point of view.
- Thoroughly understand what the customer wants to achieve.
- Reveal significant factors, at the starting, it can impact the result of the project.

Access situation:

- It requires a more detailed analysis of facts about all the resources, constraints, assumptions, and others that ought to be considered.

Determine data mining goals:

- A business goal states the target of the business terminology. For example, increase catalog sales to the existing customer.
- A data mining goal describes the project objectives. For example, It assumes how many objects a customer will buy, given their demographics details (Age, Salary, and City) and the price of the item over the past three years.

Produce a project plan:

- It states the targeted plan to accomplish the business and data mining plan.
- The project plan should define the expected set of steps to be performed during the rest of the project, including the latest technique and better selection of tools.

2. Data Understanding:

Data understanding starts with an original data collection and proceeds with operations to get familiar with the data, to data quality issues, to find better insight in data, or to detect interesting subsets for concealed information hypothesis.

Tasks:

- Collects initial data
- Describe data
- Explore data
- Verify data quality

Collect initial data:

- It acquires the information mentioned in the project resources.
- It includes data loading if needed for data understanding.
- It may lead to original data preparation steps.
- If various information sources are acquired then integration is an extra issue, either here or at the subsequent stage of data preparation.

Describe data:

- It examines the "gross" or "surface" characteristics of the information obtained.
- It reports on the outcomes.

Explore data:

- Addressing data mining issues that can be resolved by **querying, visualizing,** and **reporting,** including:
  - Distribution of important characteristics, results of simple aggregation.
  - Establish the relationship between the small number of attributes.
  - Characteristics of important sub-populations, simple statical analysis.
- It may refine the data mining objectives.
- It may contribute or refine the information description, and quality reports.
- It may feed into the transformation and other necessary information preparation.

Verify data quality:

- It examines the data quality and addressing questions.

3. Data Preparation:

- It usually takes more than 90 percent of the time.
- It covers all operations to build the final data set from the original raw information.
- Data preparation is probable to be done several times and not in any prescribed order.

Tasks:

- Select data
- Clean data
- Construct data
- Integrate data

- Format data

Select data:

- It decides which information to be used for evaluation.
- In the data selection criteria include significance to data mining objectives, quality and technical limitations such as data volume boundaries or data types.
- It covers the selection of characteristics and the choice of the document in the table.

Clean data:

- It may involve the selection of clean subsets of data, inserting appropriate defaults or more ambitious methods, such as estimating missing information by modeling.

Construct data:

- It comprises of Constructive information preparation, such as generating derived characteristics, complete new documents, or transformed values of current characteristics.

Integrate data:

- Integrate data refers to the methods whereby data is combined from various tables, or documents to create new documents or values.

Format data:

- Formatting data refer mainly to linguistic changes produced to information that does not alter their significance but may require a modeling tool.

4. Modeling:

In modeling, various modeling methods are selected and applied, and their parameters are measured to optimum values. Some methods gave particular requirements on the form of data. Therefore, stepping back to the data preparation phase is necessary.

Tasks:

- Select modeling technique
- Generate test design
- Build model
- Access model

Select modeling technique:

- It selects the real modeling method that is to be used. For example, decision tree, neural network.
- If various methods are applied,then it performs this task individually for each method.

Generate test Design:

- Generate a procedure or mechanism for testing the validity and quality of the model before constructing a model. For example, in classification, error rates are commonly used as quality measures for data mining models. Therefore, typically separate the data set into train and test set, build the model on the train set and assess its quality on the separate test set.

Build model:

- To create one or more models, we need to run the modeling tool on the prepared data set.

Assess model:

- It interprets the models according to its domain expertise, the data mining success criteria, and the required design.
- It assesses the success of the application of modeling and discovers methods more technically.
- It Contacts business analytics and domain specialists later to discuss the outcomes of data mining in the business context.

5. Evaluation:

- At the last of this phase, a decision on the use of the data mining results should be reached.
- It evaluates the model efficiently, and review the steps executed to build the model and to ensure that the business objectives are properly achieved.
- The main objective of the evaluation is to determine some significant business issue that has not been regarded adequately.
- At the last of this phase, a decision on the use of the data mining outcomes should be reached.

Tasks:

- Evaluate results
- Review process
- Determine next steps

Evaluate results:

- It assesses the degree to which the model meets the organization's business objectives.
- It tests the model on test apps in the actual implementation when time and budget limitations permit and also assesses other data mining results produced.
- It unveils additional difficulties, suggestions, or information for future instructions.

Review process:

- The review process does a more detailed evaluation of the data mining engagement to determine when there is a significant factor or task that has been somehow ignored.
- It reviews quality assurance problems.

Determine next steps:

- It decides how to proceed at this stage.
- It decides whether to complete the project and move on to deployment when necessary or whether to initiate further iterations or set up new data-mining initiatives.it includes resources analysis and budget that influence the decisions.

6. Deployment:

Determine:

- Deployment refers to how the outcomes need to be utilized.

Deploy data mining results by:

- It includes scoring a database, utilizing results as company guidelines, interactive internet scoring.
- The information acquired will need to be organized and presented in a way that can be used by the client. However, the deployment phase can be as easy as producing. However, depending on the demands, the deployment phase may be as simple as generating a report or as complicated as applying a repeatable data mining method across the organizations.

Tasks:

- Plan deployment
- Plan monitoring and maintenance
- Produce final report
- Review project

Plan deployment:

- To deploy the data mining outcomes into the business, takes the assessment results and concludes a strategy for deployment.
- It refers to documentation of the process for later deployment.

Plan monitoring and maintenance:

- It is important when the data mining results become part of the day-to-day business and its environment.
- It helps to avoid unnecessarily long periods of misuse of data mining results.
- It needs a detailed analysis of the monitoring process.

Produce final report:

- A final report can be drawn up by the project leader and his team.
- It may only be a summary of the project and its experience.
- It may be a final and comprehensive presentation of data mining.

Review project:

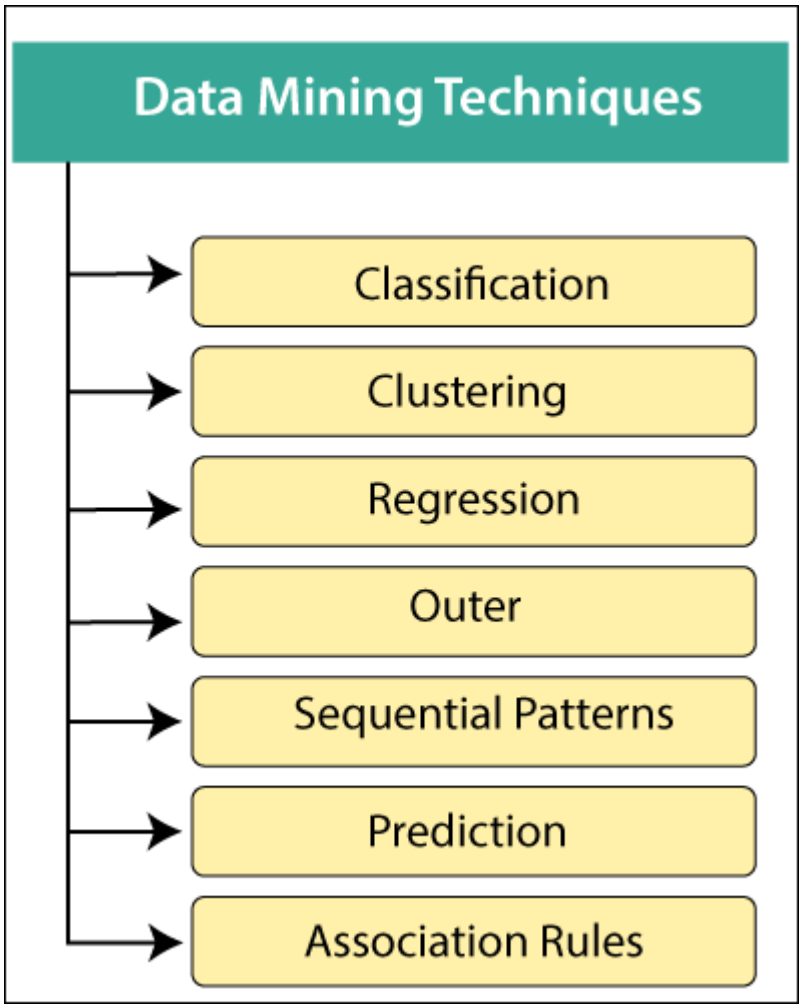
- Review projects evaluate what went right and what went wrong, what was done wrong, and what needs to be improved.

Data Mining Techniques

Data mining includes the utilization of refined data analysis tools to find previously unknown, valid patterns and relationships in huge data sets. These tools can incorporate statistical models, machine learning techniques, and mathematical algorithms, such as neural networks or decision trees. Thus, data mining incorporates analysis and prediction.

Depending on various methods and technologies from the intersection of machine learning, database management, and statistics, professionals in data mining have devoted their careers to better understanding how to process and make conclusions from the huge amount of data, but what are the methods they use to make it happen?

In recent data mining projects, various major data mining techniques have been developed and used, including association, classification, clustering, prediction, sequential patterns, and regression.



1. Classification:

This technique is used to obtain important and relevant information about data and metadata. This data mining technique helps to classify data in different classes.

Data mining techniques can be classified by different criteria, as follows:

i. **Classification of Data mining frameworks as per the type of data sources mined:**
   This classification is as per the type of data handled. For example, multimedia, spatial data, text data, time-series data, World Wide Web, and so on..

ii. **Classification of data mining frameworks as per the database involved:**
   This classification based on the data model involved. For example. Object-oriented database, transactional database, relational database, and so on..

iii. **Classification of data mining frameworks as per the kind of knowledge discovered:**
   This classification depends on the types of knowledge discovered or data mining functionalities. For example, discrimination, classification, clustering, characterization, etc. some frameworks tend to be extensive frameworks offering a few data mining functionalities together..

iv. **Classification of data mining frameworks according to data mining techniques used:**
   This classification is as per the data analysis approach utilized, such as neural networks, machine learning, genetic algorithms, visualization, statistics, data warehouse-oriented or database-oriented, etc.
   The classification can also take into account, the level of user interaction involved in the data mining procedure, such as query-driven systems, autonomous systems, or interactive exploratory systems.

2. Clustering:

Clustering is a division of information into groups of connected objects. Describing the data by a few clusters mainly loses certain confine details, but accomplishes improvement. It models data by its clusters. Data modeling puts clustering from a historical point of view rooted in statistics, mathematics, and numerical analysis. From a machine learning point of view, clusters relate to hidden patterns, the search for clusters is unsupervised learning, and the subsequent framework represents a data concept. From a practical point of view, clustering plays an extraordinary job in data mining applications. For example, scientific data exploration, text mining, information retrieval, spatial database applications, CRM, Web analysis, computational biology, medical diagnostics, and much more.

In other words, we can say that Clustering analysis is a data mining technique to identify similar data. This technique helps to recognize the differences and similarities between the

data. Clustering is very similar to the classification, but it involves grouping chunks of data together based on their similarities.

3. Regression:

Regression analysis is the data mining process is used to identify and analyze the relationship between variables because of the presence of the other factor. It is used to define the probability of the specific variable. Regression, primarily a form of planning and modeling. For example, we might use it to project certain costs, depending on other factors such as availability, consumer demand, and competition. Primarily it gives the exact relationship between two or more variables in the given data set.

4. Association Rules:

This data mining technique helps to discover a link between two or more items. It finds a hidden pattern in the data set.

Association rules are if-then statements that support to show the probability of interactions between data items within large data sets in different types of databases. Association rule mining has several applications and is commonly used to help sales correlations in data or medical data sets.

The way the algorithm works is that you have various data, For example, a list of grocery items that you have been buying for the last six months. It calculates a percentage of items being purchased together.

These are three major measurements technique:

- **Lift:**
  This measurement technique measures the accuracy of the confidence over how often item B is purchased.
  (Confidence) / (item B)/ (Entire dataset)
- **Support:**
  This measurement technique measures how often multiple items are purchased and compared it to the overall dataset.
  (Item A + Item B) / (Entire dataset)
- **Confidence:**
  This measurement technique measures how often item B is purchased when item A is purchased as well.
  (Item A + Item B)/ (Item A)

5. Outer detection:

This type of data mining technique relates to the observation of data items in the data set, which do not match an expected pattern or expected behavior. This technique may be used in various domains like intrusion, detection, fraud detection, etc. It is also known as Outlier

Analysis or Outilier mining. The outlier is a data point that diverges too much from the rest of the dataset. The majority of the real-world datasets have an outlier. Outlier detection plays a significant role in the data mining field. Outlier detection is valuable in numerous fields like network interruption identification, credit or debit card fraud detection, detecting outlying in wireless sensor network data, etc.

6. Sequential Patterns:

The sequential pattern is a data mining technique specialized for **evaluating sequential data** to discover sequential patterns. It comprises of finding interesting subsequences in a set of sequences, where the stake of a sequence can be measured in terms of different criteria like length, occurrence frequency, etc.

In other words, this technique of data mining helps to discover or recognize similar patterns in transaction data over some time.

7. Prediction:

Prediction used a combination of other data mining techniques such as trends, clustering, classification, etc. It analyzes past events or instances in the right sequence to predict a future event.